DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
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ACCESSVR
VR WHEELCHAIR CONTROLLER

JAKE COOPER
ERIC NGUYEN
KHOA NGUYEN
SANTIAGO HERNANDEZ

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1 Product Concept

This section outlines the concept for our Virtual Reality (VR) wheelchair system, detailing its purpose, use, and target audience. The VR wheelchair is designed to enhance the training and navigation skills of wheelchair users by providing a realistic and immersive virtual environment. Users will be able to practice various scenarios and movements in a safe and controlled virtual space, thereby improving their confidence and skills in real-world wheelchair navigation.

1.1 Purpose and Use

The VR wheelchair system aims to provide an innovative training platform for wheelchair users to practice and improve their navigation skills within a virtual reality environment. By simulating real-world scenarios and obstacles, the system allows users to experience and adapt to various challenges without physical risk. This tool is intended to be used in rehabilitation centers, training facilities, and by individuals at home who seek to enhance their wheelchair maneuvering abilities. The system integrates with a physical wheelchair to provide a realistic and responsive experience that mimics actual movement and interaction within virtual settings.

1.2 Intended Audience

The intended audience for the VR wheelchair system includes individuals who use wheelchairs, whether due to temporary or permanent mobility impairments. This product is designed for both new wheelchair users who need to develop basic navigation skills and experienced users who wish to refine and expand their capabilities. Additionally, rehabilitation centers, physical therapy clinics, and organizations providing support for wheelchair users can utilize this system for training and educational purposes. The system is adaptable to various user needs and skill levels, making it a versatile tool for a wide range of applications in both individual and institutional contexts.



Figure 1: VR wheelchair conceptual drawing

2 Product Description

This section provides the reader with an overview of the VR Wheelchair System. It details the primary operational aspects of the product, focusing on the experiences and interactions of end users, the responsibilities of maintainers, and the roles of administrators. The key features and functions of the system, including critical user interactions and interfaces, are described in detail to ensure a comprehensive understanding of the product's capabilities.

2.1 Features & Functions

The VR Wheelchair System is designed to transform standard long-term wheelchairs into controllers for navigating virtual reality (VR) environments. The system includes several key features and functions:

- Virtual Environments: The system includes a variety of carefully designed VR environments that simulate different real-life scenarios. These environments provide users with opportunities to practice and enhance their navigation skills in a safe and controlled setting.
- User Interface: The VR system features an intuitive user interface that allows users to easily interact with the virtual environment. The interface is designed to be accessible and user-friendly, ensuring that wheelchair users can navigate the VR world with ease.
- Sensor-Based Modules: The system utilizes sensor-based modules that attach to the wheels of the wheelchair. These sensors capture the movement of the wheels and translate it into corresponding actions within the VR environment. This allows users to control their movement in the VR world by spinning the wheels in real life.
- Customization and Adaptability: The system can be customized to meet the specific needs of individual users. This includes adjusting the sensitivity of the sensors, the difficulty of the VR environments, and other settings to ensure an optimal training experience.

Principal components:

- Wheelchair Attachments: Sensor-based modules attached to the wheels.
- VR Headset: Provides the immersive visual experience.
- Control unit: Processes the sensor data and translates it into VR control.
- VR Software: Manage the virtual environments and user interactions.

2.2 External Inputs & Outputs

The VR Wheelchair system relies on various external inputs and outputs to function effectively. These data flows are critical for the system's operation:

Name	Description	Use
Wheelchair	Wheels will replace thumbsticks for locomotion	Controls movement.
Hands	Hand-tracking through Quest 3 API	Grabbing items in VR.
VR Environment Display	Visual and auditory output through VR headset	Provides an immersive experience.

Table 2: User Input and Outputs

2.3 PRODUCT INTERFACES

This section specifies what the operational interfaces of the VR Wheelchair System will look like for end users, administrators, and maintainers. Although the interfaces are still in development, the following descriptions and mock-ups provide an overview of the intended design and functionality.

- End-User Interface
  - Main menu: Accessible via the VR headset, the main menu will allow users to select different training environments, adjust settings, and view their performance metrics.
  - Control panel: A virtual control panel will provide options to start, pause, and stop the VR session, as well as access help and support features.
- Administrator Interface
  - System Settings: Allows for adjustments to sensor sensitivity, VR environment settings, and overall system performance.
  - Reports and Analytics: Provides detailed reports on user progress and system usage statistics.
- Maintainer Interface
  - System Updates: Allows for the installation of software updates and patches.
  - Maintenance Logs: Tracks maintenance activities and system performance over time.

3 Customer Requirements

This section outlines the specific features and functions required for the VR Wheelchair System, as specified by wheelchair users, physical therapists, and other stakeholders. These requirements establish the expected "look and feel" of the product, detailing what end users should expect the system to do. Each requirement addresses a specific customer need, ensuring the system meets their expectations for training and navigation. Requirements in this section are created in collaboration with customers and must not be altered without their explicit agreement.

3.1 Locomotion Requirement

3.1.1 Description

The user will be able to move around the virtual environment using the wheelchair's wheels as the movement controller. The system will accurately capture the wheel movements and translate them into corresponding actions within the VR environment, allowing users to navigate through different terrains smoothly and intuitively.

3.1.2 Source

Sponsor.

3.1.3 Constraints

Health Safety: The system must be safe to use, with no risk of injury from the equipment or VR experience.

3.1.4 Standards

ISO 7176: Specifies the requirements and test methods for the performance of wheelchairs, including electronic components [1].
IEEE 11073: Standards for medical devices communication [2].

3.1.5 Priority

Critical.

3.2 Interface Requirement

3.2.1 Description

The VR Wheelchair System will include a user-friendly interface that allows users to easily navigate the VR environment, access settings, and view performance metrics. The interface should be intuitive, with clear instructions and feedback.

3.2.2 Source

Sponsor.

3.2.3 Constraints

Technical: The interface must be compatible with the hardware components and VR headset used in the system.
Ethical: The interface should be designed to be accessible to users with disabilities, adhering to principles of universal design.

3.2.4 Standards

ISO 9241-11: Ergonomics of human system interaction. Guidance on usability [3].

3.2.5 PRIORITY

High.

4 Packaging Requirements

The VR wheelchair is primarily being developed for rehabilitation and physical therapy. However, the packaging requirements would be very similar whether or it would be commercialized. This product includes both hardware and software components.

4.1 Hardware Component

4.1.1 Description

All hardware components required for the wheel tracking including an encoder and microcontroller that can seamlessly attach to the a wheelchair. These should be securely packaged together in a single package. However, in situations where components are not pre-assembled, instructions must be provided for setup by the customer.

4.1.2 Source

Senior Design Team - VR Wheelchair

4.1.3 Constraints

Must be safe and not cause any injury

4.1.4 Standards

Must follow ADA guidelines

4.1.5 Priority

High

4.2 Software Component

4.2.1 Description

The testing environment will be developed to be used in a Meta Quest headset, so must be able to accessed through the regular distribution channels of the Meta Quest. If this is not possible, it will be in a some sort of portable memory device to be loaded onto the Meta Quest manually.

4.2.2 Source

Senior Design Team - VR Wheelchair

4.2.3 Constraints

Must the program must be within the memory contraints of the microprocessor that is being used

4.2.4 Standards

The package should follow HIPAA and ISO to ensure a safe and high quality project

4.2.5 Priority

High

5 Performance Requirements

This section will expand on the requirements needed for the performance of the wheelchair controller and the in-game experience for the user. Interaction between the controller and the Meta Quest 3 headset, the nature of the virtual environment, and providing power to the system are all examined.

5.1 Controller Response

5.1.1 Description

The wheelchair controller should provide accurate, real-time response in the virtual environment. The speed and direction of wheel movement should be sent to the Quest 3 headset and provide feedback in the environment in a way that is commensurate with the behavior of a wheelchair operating in the real world. The response time from user input to virtual feedback should be on par with other similar controllers, between 10 and 20 ms.

5.1.2 Source

Shawn Gieser

5.1.3 Constraints

Especially due to the product's nature as a simulation, the user expects virtual movement that feels accurate to the amount of input they are giving. The direction and speed of the wheelchair in the game should reflect the user's own manual manipulation of the wheelchair controller's wheels, with a small margin of error.

5.1.4 Standards

No applicable standards.

5.1.5 Priority

Critical.

5.2 System Power

5.2.1 Description

The Meta Quest headset is powered by its own internal, rechargeable battery. The wheelchair controller requires a minimum of 5V for the rotary encoders, and a minimum of 3.3V to power the microcontroller. Due to the large power draw, powering the wheelchair controller requires a sacrifice of either portability or power time. For this project, the controller will be powered with a transformer plugged directly into a wall outlet, thereby sacrificing portability in the name of longer power.

5.2.2 Source

Shawn Gieser and AccessVR.

5.2.3 Constraints

As this device is most likely to be used as a tool for occupational therapy, or set up in a single spot to allow for individual participation, users would reasonably expect to get extended use from the device without worrying about battery charge. Therefore, having the controller plugged directly into an outlet seems most appropriate to provide the longest user experience, at the cost of some measure of portability.

5.2.4 Standards

16 CFR 1505.6

5.2.5 Priority

Critical.

5.3 Virtual Simulation

5.3.1 Description

The virtual simulation within the Quest 3 headset should provide an accurate simulation of situations experienced by everyday wheelchair users. The simulation should be geared toward the hardware and software capabilities of the Quest 3 headset, and should run smoothly with reasonable load and response times.

5.3.2 Source

AccessVR

5.3.3 Constraints

When utilizing this product, the user expects to experience life as one who uses a wheelchair every day would. This means providing the user with true-to-life situations and environments, and doing so at minimal inconvenience to the user by creating a simulation that is reasonably free of tearing, stuttering, and slow load times.

5.3.4 Standards

No applicable standards.

5.3.5 Priority

High

6 Safety Requirements

Include a header paragraph specific to your product here. Safety requirements might address items specific to your product such as: no exposure to toxic chemicals; lack of sharp edges that could harm a user; no breakable glass in the enclosure; no direct eye exposure to infrared/laser beams; packaging/grounding of electrical connections to avoid shock; etc.

6.1 Laboratory equipment lockout/tagout (LOTO) procedures

6.1.1 Description

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

6.1.2 Source

CSE Senior Design laboratory policy

6.1.3 Constraints

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 Standards

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 Priority

Critical

6.2 National Electric Code (NEC) wiring compliance

6.2.1 Description

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

6.2.2 Source

CSE Senior Design laboratory policy

6.2.3 Constraints

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

6.2.4 Standards

NFPA 70

6.2.5 Priority

Critical

6.3 RIA robotic manipulator safety standards

6.3.1 Description

Robotic manipulators, if used, will either housed in a compliant lockout cell with all required safety interlocks, or certified as a "collaborative" unit from the manufacturer.

6.3.2 Source

CSE Senior Design laboratory policy

6.3.3 Constraints

Collaborative robotic manipulators will be preferred over non-collaborative units in order to minimize potential hazards. Sourcing and use of any required safety interlock mechanisms will be the responsibility of the engineering team.

6.3.4 Standards

ANSI/RIA R15.06-2012 American National Standard for Industrial Robots and Robot Systems, RIA TR15.606-2016 Collaborative Robots

6.3.5 Priority

Critical

6.4 Wheelchair Stability

6.4.1 Description

In order to be used as a controller for the simulation, the rear wheels of the wheelchair must be elevated off the ground to allow for free movement. As the user must also be sitting in the wheelchair while the rear wheels are elevated, the means of lifting the wheelchair off the ground must be stable and sturdy, able to withstand a large amount of user movement.

6.4.2 Source

Shawn Gieser and AccessVR

6.4.3 Constraints

The wheelchair cannot be effectively and safely used as a controller while on the ground. Since the back wheels must be elevated while the user is climbing into and out of the chair, and spinning the back wheels to control the simulation, this also presents safety concerns. Elevation of the wheelchair must be effective and allow for free movement of the rear wheels, but also very stable and completely safe for public use.

6.4.4 Priority

Critical

7 Maintenance & Support Requirements

Maintenance and support requirements are essential, especially in a healthcare setting. As the developers of this solution, we envision the following needs:

7.1 Technical Support

7.1.1 Description

A dedicated support line will be available to address issues that may arise after production

7.1.2 Source

Senior Design Team - VR Wheelchair

7.1.3 Constraints

Support most likely will not be available 24/7

7.1.4 Standards

General customer serice standards present in the U.S.

7.1.5 Priority

Medium

7.2 Hardware Maintenance

7.2.1 Description

As hardware slowly degrades overtime, there may be a need to repair or replace parts over time

7.2.2 Source

Senior Design Team - VR Wheelchair

7.2.3 Constraints

Cost and part availability

7.2.4 Standards

General customer serice standards present in the U.S.

7.2.5 Priority

Medium

8 Other Requirements

The additional requirements of the VR Wheelchair aim to encompass crucial additions that can be implemented to greatly improve the experience of the rehabilitation and ultimately expedite the learning process.

8.1 Software Licensing

8.1.1 Description

Third-party software components or libraries utilized must possess clear licensing that permits commercial use without any infringements.

8.1.2 Source

Legal and Compliance Department

8.1.3 Constraints

Cost associated with these commercial licenses

8.1.4 Standards

Open Source Initiative (OSI) Licensing Standards

8.1.5 Priority

High

8.2 Product Scalability

8.2.1 Description

The system should be designed in a way that allows for future items mentioned in the next sectio such as integrating motors and a tilt plate. RFID readers or adding additional features.

8.2.2 Source

Senior Design Team - VR Wheelchair

8.2.3 Constraints

Must fall within our allocated budget and not conflict with previously implemented hardware.

8.2.4 Standards

ISO/IEC 9126 Software engineering

8.2.5 Priority

Medium

9 Future Items

Our primary objective is to ensure the wheels are properly tracked to achieve a functional product. Meanwhile, these upcoming features will significantly enhance the user experience, making it more realistic and interactive for players. Ultimately, this should expedite the learning process.

9.1 Resistance

9.1.1 Description

Different floor textures would have different resistance coefficients which could be implemented via a brakes. An example is that grass is much more difficult to roll in than concrete. Additionally, a motor could be implemented to show extra force needed to go up tilted environments.

9.1.2 Source

Shawn N. Gieser

9.1.3 Constraints

Must be safe and not cause any injury

9.1.4 Standards

Must follow ADA guidelines

9.1.5 Priority

Medium

9.2 Tilt Plate

9.2.1 Description

A device under the wheelchair would suspend the wheelchair up and tilt the chair in the angle that corresponds with the platform they are on in the VR environment.

9.2.2 Source

Shawn N. Gieser

9.2.3 Constraints

Must be safe and not cause any injury

9.2.4 Standards

Must follow ADA guidelines

9.2.5 Priority

Medium

REFERENCES

[1] BSI Group. Multi-part document bs iso 7176 - wheelchairs.

[2] IEEE. Ieee standard for health informatics - device interoperability - part 10701: Point-of-care medical device communication - metric provisioning by participants in a service-oriented device connectivity (sdc) system.

[3] ISO. Iso 9241-11:2018(en) ergonomics of human-system interaction â part 11: Usability: Definitions and concepts.